



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Charge and Current — $Q = It$

One equation, one trap, and one number that puts the whole thing in perspective.

Key concept

CURRENT IS A FLOW OF CHARGE

Current is not “electricity”. It is the **rate at which charge flows** — how much charge goes past a point each second.

$$\text{charge flow} = \text{current} \times \text{time} \quad Q = It$$

Q = charge flow, in **coulombs (C)** · I = current, in **amps (A)** · t = time, in **seconds (s)**

Rearranged: $I = \frac{Q}{t}$ and $t = \frac{Q}{I}$

So how much is one coulomb?

The charges doing the flowing in a metal wire are **electrons**, and a single electron carries a fantastically small charge:

$$e = 1.6 \times 10^{-19} \text{ C}$$

So a charge of Q coulombs is made up of

$$\text{number of electrons} = \frac{Q}{e}$$

This is worth doing once, because otherwise a coulomb is just a word. The numbers on this sheet — 34 C, 108 C — sound modest. They are not: 45 C is about 2.8×10^{20} electrons, or 280 million million million of them. *You will not be asked for the value of e in a Combined Science exam.*

And how fast do the electrons move?

Slowly. In a copper wire carrying a normal current the electrons **drift** along at well under a millimetre per second — slower than a snail.

Which leaves a puzzle, and the last question asks you to solve it: **if the electrons crawl, why does a lamp light the *instant* you close the switch?**

The trap, and it is the one that costs marks. t must be in **seconds**. Real questions give you *minutes*, or hours. Convert **first**, and write the converted number down so a marker can see it.

$$1 \text{ minute} = 60 \text{ s} \quad \cdot \quad 2 \text{ minutes} = 120 \text{ s} \quad \cdot \quad 1 \text{ hour} = 3600 \text{ s}$$

Worked example

A current of 0.25 A flows for **3 minutes**. Calculate the charge that flows, and the number of electrons that represents.

Convert first. $t = 3 \times 60 = 180 \text{ s}$ $Q = It = 0.25 \times 180 = \mathbf{45 \text{ C}}$

Then the electrons. $\frac{Q}{e} = \frac{45}{1.6 \times 10^{-19}} = \mathbf{2.8 \times 10^{20} \text{ electrons}}$

Using $t = 3$ would give 0.75 C — sixty times too small, and no method mark for a conversion you never wrote down.

1. State what is meant by *electric current*. [1]

The rate of flow of electrical charge.

2. A doorbell draws a current of 0.85 A for 40 seconds. Calculate the charge flow. **Give your answer to 2 significant figures.** [3]

$$Q = It = 0.85 \times 40 = 34 \text{ C} = \mathbf{34 \text{ C}} \text{ (2 s.f.)}$$

3. A phone charger supplies a current of 0.90 A for **2 minutes**. Calculate the charge that flows. [3]

$$t = 2 \times 60 = 120 \text{ s.} \quad Q = 0.90 \times 120 = \mathbf{108 \text{ C}}$$

4. A torch bulb carries a current of 0.24 A. How long must it be switched on for 72 C of charge to flow? Give your answer **in minutes**. [3]

$$t = Q/I = 72/0.24 = 300 \text{ s} = \mathbf{5 \text{ minutes.}}$$
 Convert back — the question asked for minutes.

5. A torch bulb carries a current of 0.30 A for 60 seconds. Calculate the charge that flows, and then the number of electrons. [3]

$$Q = It = 0.30 \times 60 = 18 \text{ C.} \quad \text{number} = 18/(1.6 \times 10^{-19}) = \mathbf{1.1 \times 10^{20} \text{ electrons.}}$$

6. A single closed loop carries a current of 2.0 A. Explain why the same charge flows past every point in that loop in one minute. [2]

The current is the same at every point in a single closed loop, and the time is the same, so $Q = It$ gives the same value everywhere. Charge is not used up as it goes round.

7. Electrons in a copper wire drift along at less than a millimetre per second. Explain why a lamp lights immediately when you close the switch. [3]

The wire is already full of free electrons along its whole length, including the ones inside the lamp. Closing the switch sets up the effect of the potential difference along the whole circuit almost instantly, so every electron everywhere starts to drift at essentially the same moment — including those already in the lamp. Nothing has to travel from the cell to the lamp first. Think of a straw already full of peas: push one in, one drops out of the far end at once, and no single pea travelled the straw.